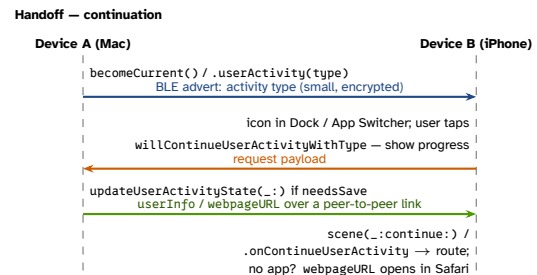
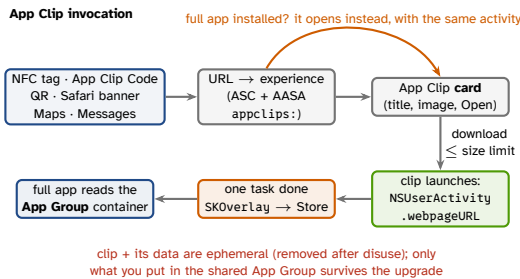


In one line: Both ride on `NSUserActivity`. An **App Clip** is a small, separately built slice of your app that a **URL** (from a code, tag, link or map) launches *without installing the app*; **Handoff** moves *your own* current activity to *your own* nearby device (same iCloud account, BLE + Wi-Fi). Neither is a deep link, and neither is state restoration.



App Clips

- **Target:** own bundle id (parent id + suffix, e.g. `.Clip`), shipped with the full app; share code via packages, strip the rest with your own `#if APPCLIP` compilation condition.
- **Size** (uncompressed binary): **10 MB** (up to iOS 15), **15 MB** (iOS 16), **100 MB** (iOS 17+ *only if* digital invocations only — no App Clip Code, NFC, QR — *and* no iOS 16 support; an App Store Connect demo link gets 100 MB with physical codes). Third-party SDKs eat it.
- **Experiences** (App Store Connect): default (one card) + **advanced** per URL prefix / location. Domain proof: `appclips:` associated domain + AASA. Launch: `webpageURL` of a `NSUserActivityTypeBrowsingWeb` activity = your routing key.
- **Ephemeral notifications:** `NSAppClipRequestEphemeralUserNotification` → up to **8 hours** without a prompt. `NSAppClipRequestLocationConfirmation` + `APActivationPayload.confirmAcquired(in:)` verifies the user is really at the code.
- **Upgrade:** `SKOverlay.AppClipConfiguration` (`.appStoreOverlay` in SwiftUI); the full app inherits only the **App Group** container. Test: `_XCAppClipURL` scheme variable; on device, Settings › Developer › Local Experiences.

Handoff & Continuity

- **Advertise:** `activityType` listed in `NSUserActivityTypes`; small plist `userInfo` (+ `requiredUserInfoKeys`), `title`, `webpageURL`; `isEligibleForHandoff` (also `ForSearch`, `ForPrediction`). Current via `becomeCurrent()` or a responder's `userActivity`; refresh lazily with `needsSave` + `updateUserActivityState(·)`.
- **Requirements:** same Apple ID / iCloud, BLE + Wi-Fi, Handoff on, same **Team ID** and activity type in both apps. Big data: continuation streams (supports `ContinuationStreams`) or the cloud.
- **Continue** per *scene*: cold start → `connectionOptions.userActivities`; running → `scene(·:continue:)`.
- **Web↔native:** app → Safari via `webpageURL`; Safari → app needs the `activitycontinuation`: associated domain (Universal Links use `applinks:`).
- **Universal Clipboard:** UIPasteboard items reach nearby devices; keep secrets local with `setItems(·:options: [localOnly: true, expirationDate: ...])`. **Continuity Camera:** iPhone as Mac camera/scanner — nothing to build on iOS.

```
ArticleView(a)
.userActivity("com.acme.article") { act in // advertise
  act.userInfo = ["id": a.id] // what, not data
  act.webpageURL = a.webpageURL // Safari fallback
  act.isEligibleForHandoff = true }
.onContinueUserActivity("com.acme.article") { act in
  if let id = act.userInfo?["id"] as? String { path.append(id) }
}
// clip AND full app: the same entry point
.onContinueUserActivity(NSUserActivityTypeBrowsingWeb) { act in
  if let url = act.webpageURL { route(url) } }
```

Four ways back into a screen

Handoff	same user, <i>other nearby device</i> , in-progress activity
Deep link	a URL anyone can send; opens the installed app (else web)
State restor.	<i>same device</i> after relaunch (<code>stateRestorationActivity</code> , <code>@SceneStorage</code>)
App Clip	a URL when the full app is <i>not</i> installed

What a clip cannot do

No background: no background `URLSession`, no Background Modes, no Bluetooth kept while unused. **Silent frameworks** (unavailable / empty / error): `AppIntents` · `BackgroundTasks` · `CallKit` · `CareKit` · `Contacts`(UI) · `CoreMotion` · `EventKit`(UI) · `FileProvider` · `HealthKit` · `HomeKit` · `MediaPlayer` · `Messages` · `MessageUI` · `NearbyInteraction` · `PhotoKit` · `ResearchKit` · `SensorKit` · `Speech`. **Privacy:** no `SKAdNetwork`, no ATT prompt; `UIDevice.name / identifierForVendor` = ""; location "When In Use" only, reset daily at 4:00; no Face ID; extensions only for Live Activity widgets; CloudKit (iOS 16+) public database read-only.

Traps

- ! Quoting "10 MB" without the OS version, or meaning the *download* size.
- ! Full app does not handle the clip's URLs — once installed, every code opens the *full* app.
- ! Clip and app share data freely — no: only an App Group; and the clip may vanish.
- ! Handoff activity never appears: type missing from `NSUserActivityTypes`, different Team ID, not current, non-plist `userInfo`, BLE off / other iCloud account.
- ! Megabytes in `userInfo`; or expecting Handoff to carry your app's login — it resumes context, not auth.
- ! Reading the pasteboard on launch — iOS 16 asks "Allow Paste"; use `UIPasteControl` / `detectPatterns`.

Remember

"Clip: a URL, one task, a slot in the App Group. Handoff: my activity, my other device, same iCloud."